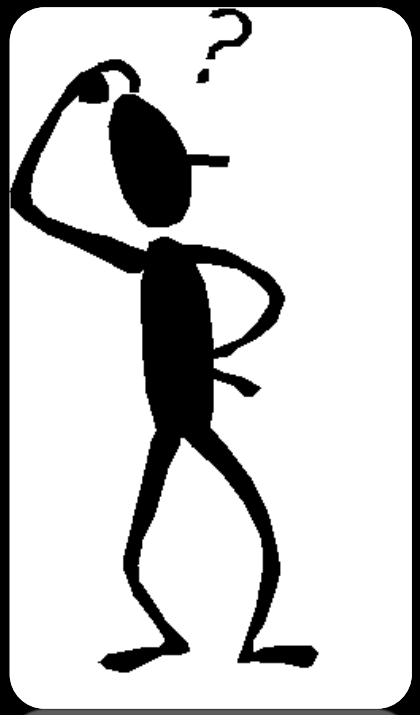
# CASE Study

The Soda Machine

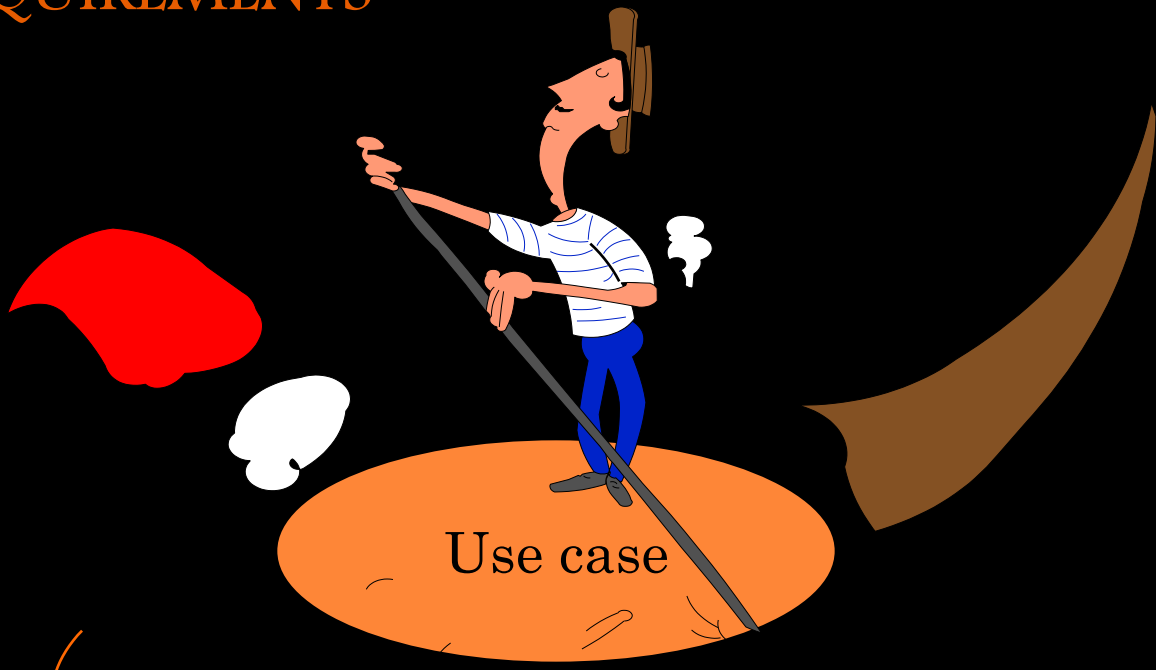
## USE CASE & STATE DIAGRAM

# Use case is important

# Why??



# USE CASES = USER REQUIREMENTS




*Way of using the system*

# Working With Use Case

- Each use case is a collection of scenarios, and each scenario is a sequence of steps. For each scenario in each use case, you'll want to show

  - The actor who initiates the use case
  - Assumptions for the scenario
  - Preconditions for the use case
  - System-related steps in the scenario
  - Postconditions when the scenario is complete
  - The actor who benefits from the use case

# Soda Machine System
## Use Cases



# BUY SODA USE CASE

## 1- Buy Soda Scenario

- The actor is a customer (who wants to purchase a can of soda)

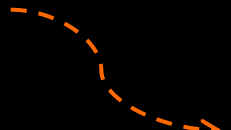
The customer initiates the scenario by

- Inserting money into the machine.
- Then makes a selection.
- If everything goes smoothly, the machine has at least one can of the selected soda in stock and presents a cold can of the soda to the customer

# Buy Soda Use Case

**1- Buy Soda Scenario (Cont)**

- Preconditions that motivate the customer (Thirst)
- Postconditions result as a consequence of the scenario's steps (customer has a soda)



What if things doesn't go smoothly??

# ALTERNATIVE PATH THROUGH THE USE CASE

- The customer doesn't have the exact amount of money the soda costs.

- Out-of-soda scenario, another sequence of steps in the "Buy soda" use case.

# BUY SODA USE CASE

## 2- Out-of-soda

### Scenario

- The customer inserts money into the machine.
- Makes a selection.
- The machine does not have at least one can of the selected soda, so it presents a message to the customer, saying it's out of that brand.
- The machine should also offer the customer the option of getting her money back.
- At this point the customer selects another brand and the machine delivers (if it's not sold out of the new selection), or takes the option of receiving the money.
- Precondition is a thirsty customer.
- Post condition is either a can of soda or the returned money.

# BUY SODA USE CASE

**3- incorrect-amount-of-money scenario**

- Let's assume the machine has the selection in stock.
- If the machine has a reserve of appropriate change on hand, it returns the difference and delivers the soda.
- If the machine doesn't have a reserve of change, it returns the money and presents a message that prompts the user for correct change.
- Precondition is the usual one.
- Post condition is either a can of soda along with change, or the returned money that was originally deposited.

# Additional Use Cases

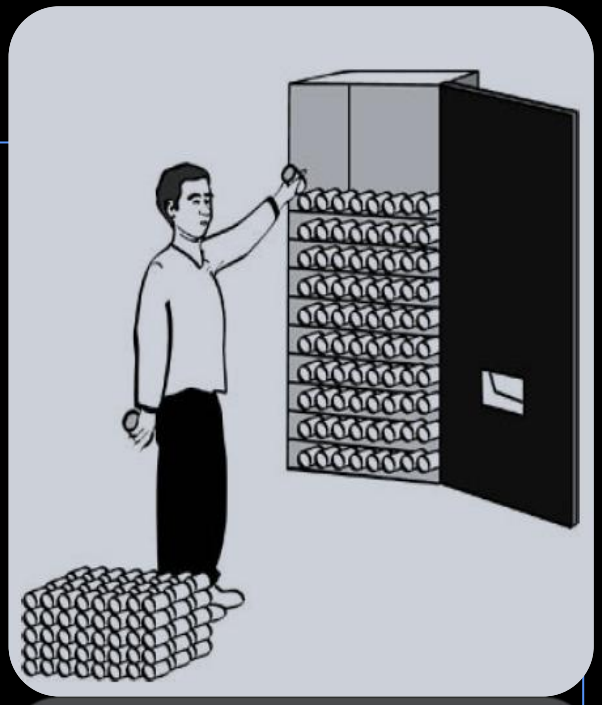
# Actors

- **The customer.**
- **Other users**

  **supplier:** restock the machine

  **collector:** collect the accumulated money from the machine

# RESTOCKING A SODA MACHINE USE CASE



- The supplier unsecure the machine.
- Pulls open the front of the machine.
- Fills each brand's compartment to capacity.
- Refills the change reserve.
- Closes the front of the machine and secures it.
- Precondition is the passage of the interval.
- Post condition is that the supplier has a new set of potential sales.

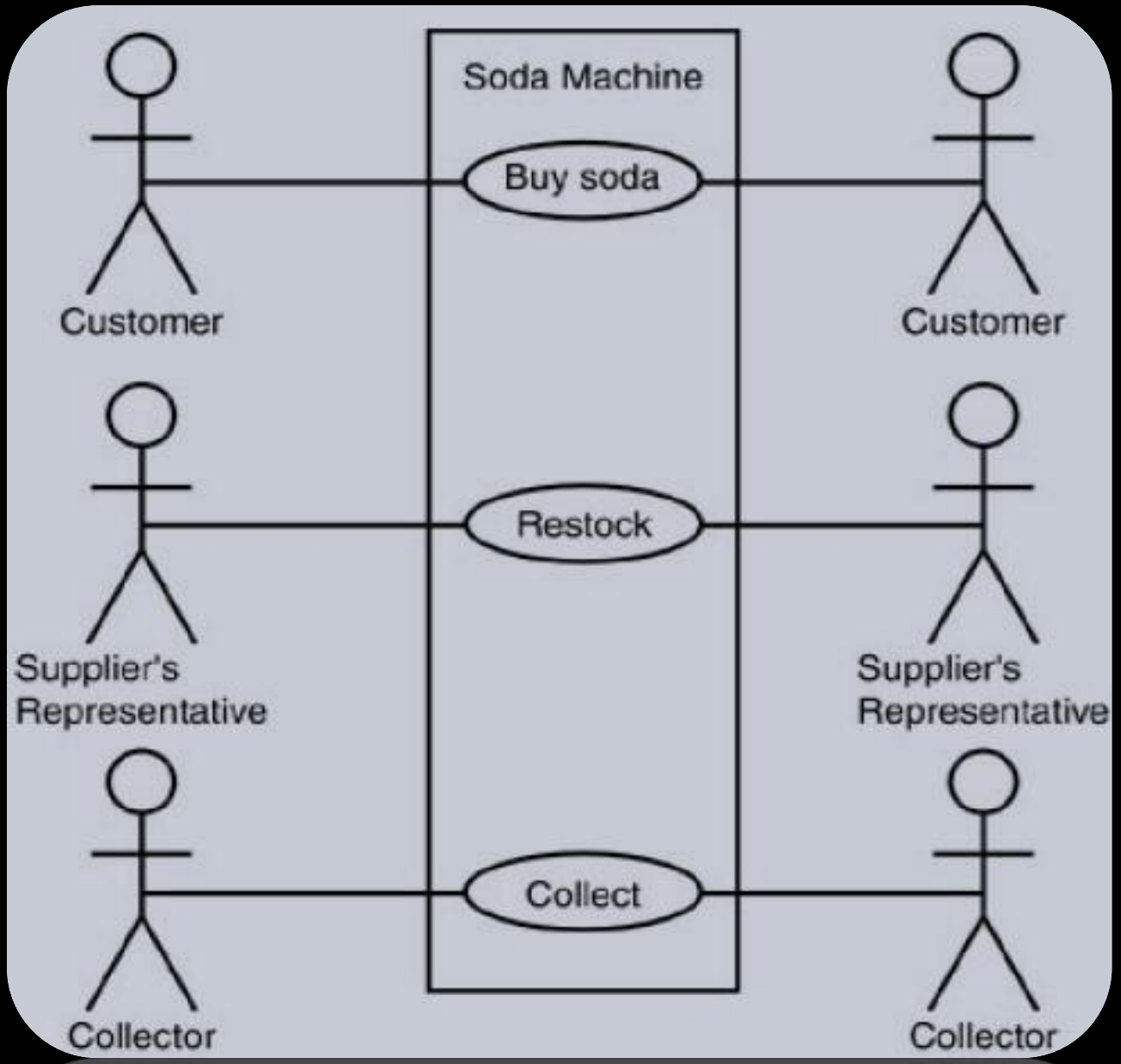
# COLLECTING THE MONEY USE CASE




Same sequence of steps as in "Restock"

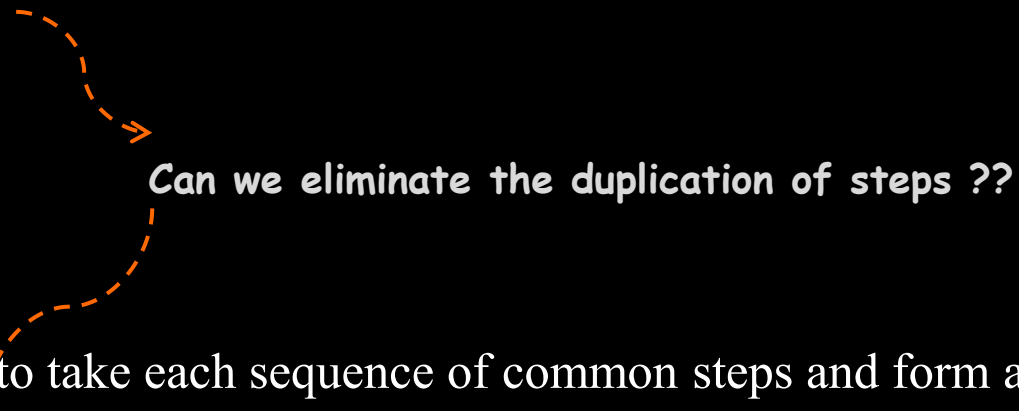
- To unsecure the machine and pull open the front.
- The collector then removes the money from the machin

  "Restock" steps of closing and securing the machine.
- Precondition is the passage of the interval.
- Postcondition is the money in the hands of the collector.


Soda Machine
Buy soda
Customer
Customer
Restock
Supplier's Representative
Supplier's Representative
Collect
Collector
Collector

# Including a Use Case

- In "Restock“ and "Collect" use case, you'll note some common steps.



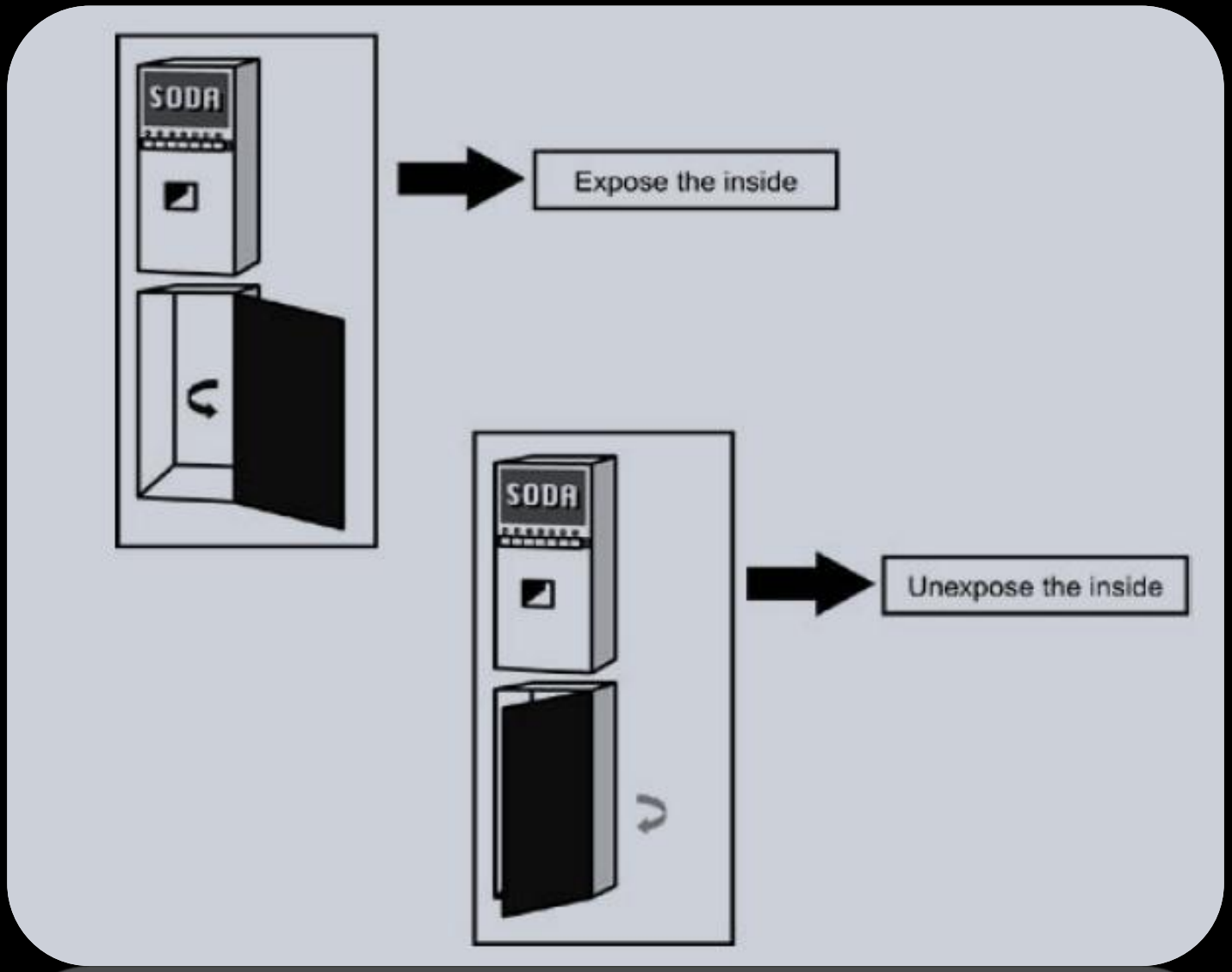

- The way to do it is to take each sequence of common steps and form an additional use case from each one.

- Combine the steps of

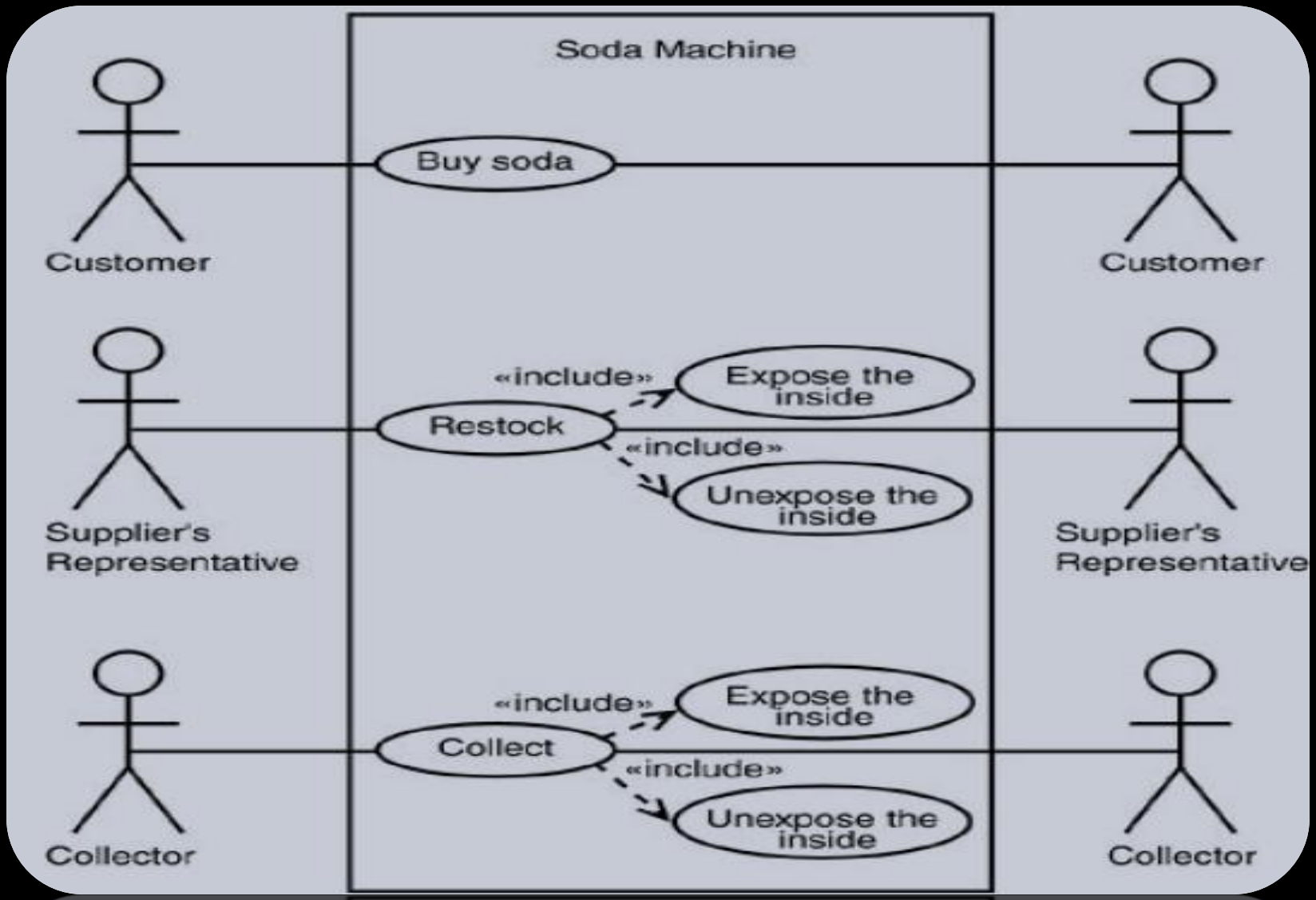
Unsecure & pull open → "Expose the inside" *use case*

Close machine & secure → "Unexpose the inside“ *use case*

"Restock" and "Collect" include the new use cases.
Accordingly, this technique of reusing a use case is referred to as **including a use case**



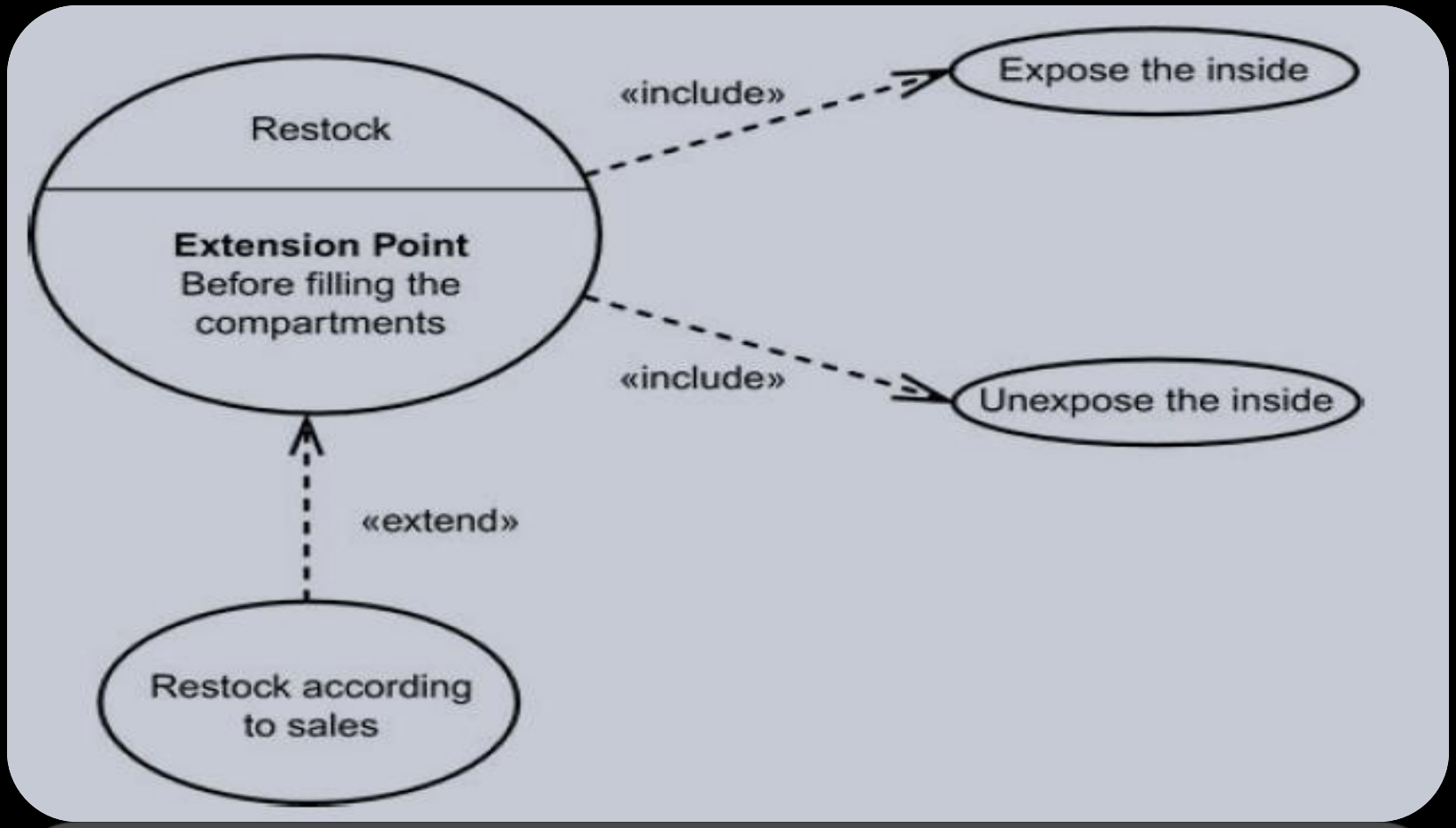
# INCLUSION


Soda Machine
Buy soda
Customer
Customer
«include»
Expose the inside
Restock
«include»
Unexpose the inside
Supplier's Representative
Supplier's Representative
«include»
Expose the inside
Collect
«include»
Unexpose the inside
Collector
Collector

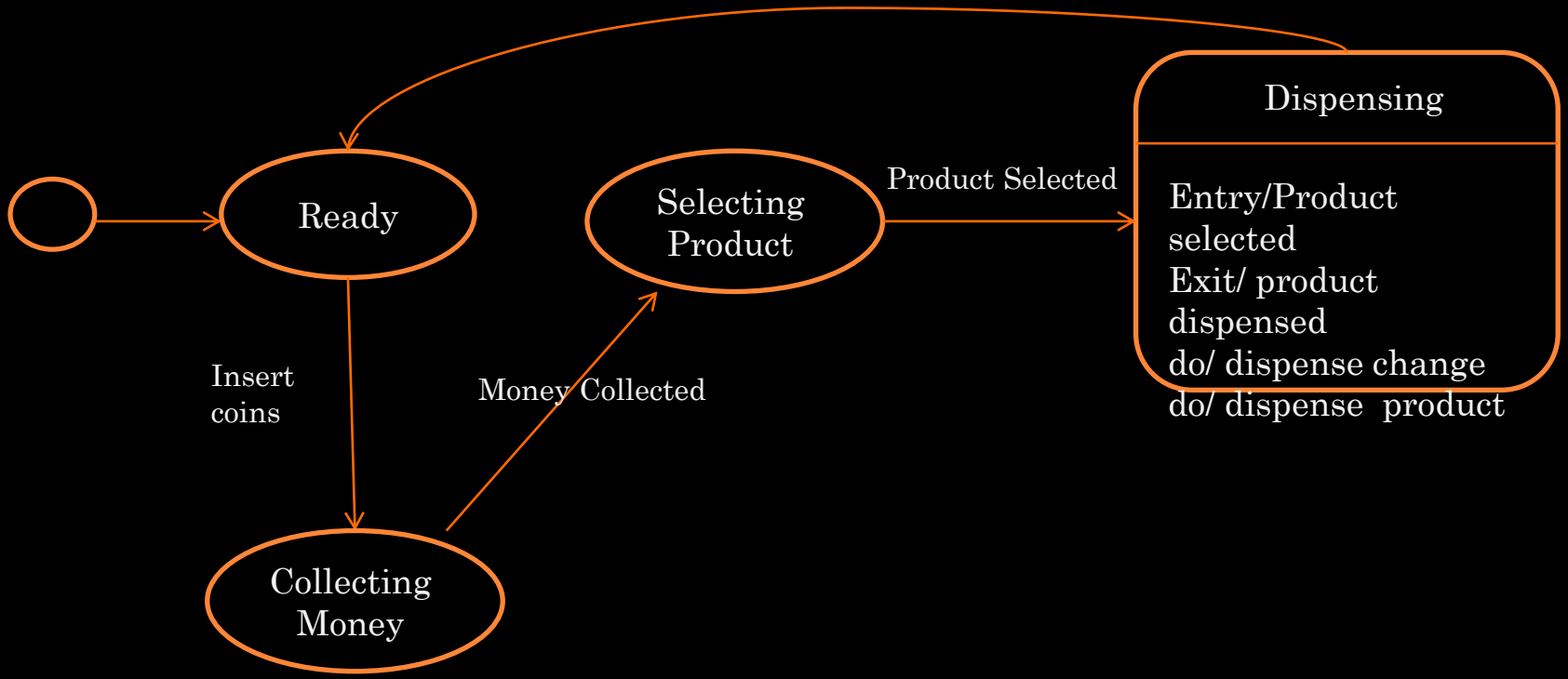
# Extending a Use Case

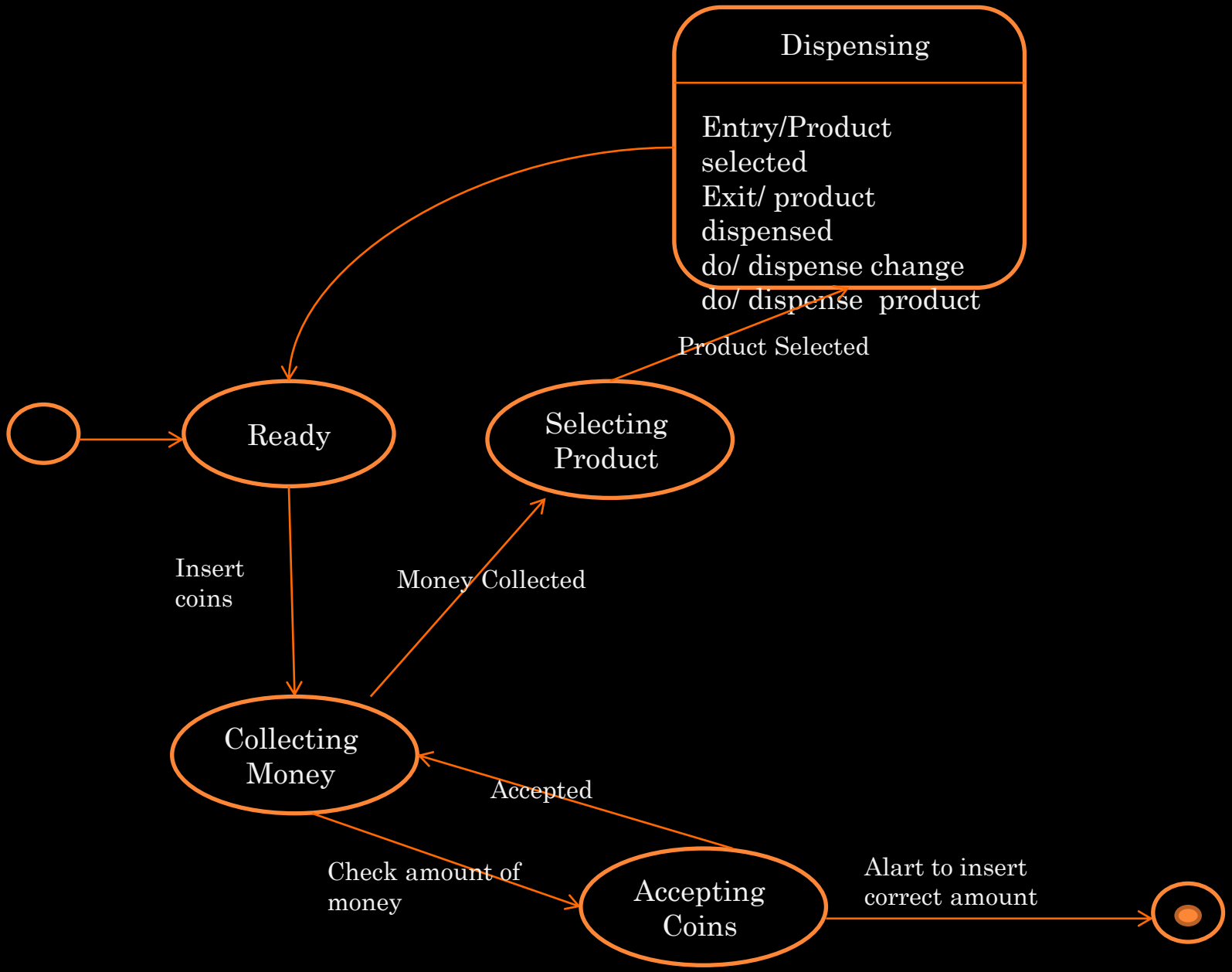
- It's possible to reuse a use case in a way other than inclusion.
- Sometimes we create a new use case by adding some steps to an existing use case.

# Extension


Restock
Extension Point
Before filling the compartments
«include»
Expose the inside
«include»
Unexpose the inside
«extend»
Restock according to sales

# State Diagram




Dispensing
Entry/Product selected
Exit/ product dispensed
do/ dispense change
do/ dispense product
Product Selected
Ready
Selecting Product
Insert coins
Money Collected
Collecting Money
Accepted
Check amount of money
Accepting Coins
Alart to insert correct amount